



CHALLENGE 03 / 20

THE GHOST ON THE SALES DIRECTOR'S LAPTOP

COVERT REMOTE-CONTROL DETECTION WITH NO PERSISTENT ARTIFACTS

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A listed manufacturer's confidential pricing, rebate structures and tender bids appeared in a competitor's hands. The leak is traced to the Sales Director's laptop, used across 14 airports, nine hotels and more than thirty client sites in 11 weeks. Endpoint protection, EDR and vulnerability scans are clean.

Four weeks ago IT reimaged the laptop as routine maintenance. The old disk is gone. What remains is the new disk, cloud and conditional-access records, VPN logs, device-management records, network-side metadata, and the Director's other devices. The Director's assistant, a travelling colleague and IT support have all legitimately used or remotely serviced the machine.

Legal must decide whether this is a disciplinary matter or a security breach. That depends on whether anyone other than the Director operated the laptop.

2 | PROBLEM STATEMENT

Develop a **Covert Session Control Detection System** that determines whether a device was operated by someone other than its legitimate user over a multi-month period, using only residual behavioural, network-side and cloud-side evidence, when no malware, no persistence mechanism and no original disk artefact survives.

The system must identify which sessions and actions were non-owner, the likely control channel, and whether the operator was a legitimate delegate or a covert party.

Hard constraints

- No malicious file or process exists, and the prior system state was destroyed by reimaging.
- The owner's own behaviour is irregular because of travel, jet lag, shared screens and delegates.
- The adversary acts in the owner's idle windows and imitates his cadence.
- Legitimate remote support and assistants must be separated from a covert operator.
- The system must be able to say the question cannot be determined, and say exactly why.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Owner Behaviour Modelling	Ability to model an irregular, travelling user without a clean baseline.
02	Non-Owner Operation Detection	Ability to detect operation by someone else inside an owner's session.
03	Control Channel Inference	Ability to infer whether control was remote, local or scripted.
04	Delegate Separation	Ability to separate sanctioned delegates and IT support from covert use.
05	Mimicry Resistance	Ability to resist an adversary imitating the owner's cadence.
06	Residual-Evidence Fusion	Ability to combine weak evidence from many independent sources.
07	Reimaging Impact Assessment	Ability to state what the reimaging made unrecoverable.
08	Undeterminable-Outcome Reporting	Ability to state when no defensible conclusion exists.

The system must clearly distinguish between:

OWNER	SECOND OPERATOR	REMOTE	LOCAL	UNDETERMINED
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Absence of malware must not be presented as absence of control.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated device history including a reimage
02	Sessions by an owner, a delegate, IT support and a covert operator
03	Per-session non-owner detection
04	Control channel inference
05	Operation against an adversary that mimics the owner
06	Separation of delegates from the covert operator
07	An explained undeterminable outcome

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.